AI & SOCIETY COURSE

Prof. Mor Naaman

Part 1

AI FOUNDATIONS

Unit 3: Data and Evaluation

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Where We Are in the Course

Part 1 — AI Foundations

1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals

3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information

5. AI-Mediated Communication and Relationships
6. AI and the Information Environment

Part 4 — Challenges for AI Implementation

7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact

10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI

12. The Future: Navigating AI in Your Community

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This lesson's plan

Open and close models

What data do these models actually use?

How are the models evaluated?

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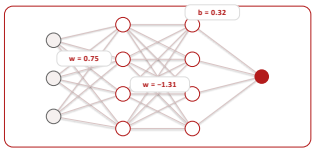
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Who has access to the model weights?

Open and close-weight models

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Remember the model parameters?



An open-weight model publishes the neural network and these weights/parameters to everyone so they can use it on their computers/servers.

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Closed- vs Open-Weight Models

Are the model's weights available to everyone?

Closed-Weight

The weights are private. You reach the model only through a service where you can use it, but not download to your computer or inspect it.

Examples: GPT-5, Claude, Gemini

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Closed- vs Open-Weight Models

Are the model's weights available to everyone?

Closed-Weight

The weights are private. You reach the model only through a service where you can use it, but not download to your computer or inspect it.

Examples: GPT-5, Claude, Gemini

Open-Weight

The trained weights are published. Anyone can download them, run the model themselves, or fine-tune it.

Examples: Llama, DeepSeek, Qwen, Mistral, Kimi

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Closed- vs Open-Weight Models

Sometimes just called "open" or "closed" models

Closed Models

The weights are private. You reach the model only through a service where you can use it, but not download to your computer or inspect it.

Examples: GPT-5, Claude, Gemini

Open Models

The trained weights are published. Anyone can download them, run the model themselves, or fine-tune it.

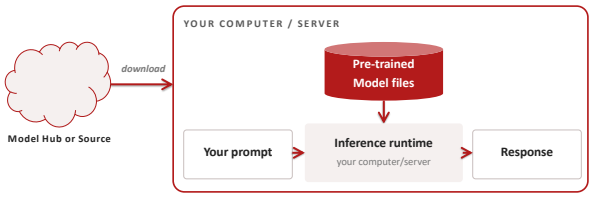
Examples: Llama, DeepSeek, Qwen, Mistral, Kimi

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Open-Weight: The Model Can Run on Any Machine

You download the model once — after that, you can do the inference on your own (or any) computer

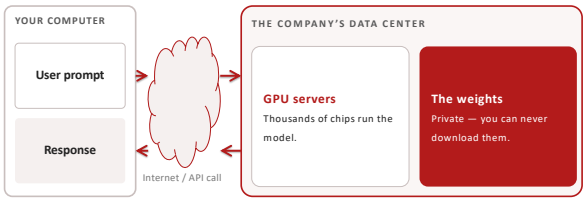


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Closed-Weight: The Model Runs on Their Servers

Your prompt goes to the server where the model runs it and sends a response.



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Open vs Closed Models: Key Differences

	Open-Weight	Closed-Weight
Where it can run	On your own (or any fitting) computer or server.	On the provider company's servers.
Cost to use	No direct cost per use (only the hardware).	Pay the provider per use (likely much more expensive).
Privacy	Your prompts never leave the machine.	Your prompts are logged, can be used by the provider.
Control	No control beyond model training.	Publisher has control: can screen prompts, change model, add restrictions.
Can you change it?	Yes: fine-tune or modify the model.	No: you get the model as the provider ships it.
What you need	A big enough computer: GPU, memory, storage.	An internet connection and an API key.
Examples	Llama, DeepSeek, Qwen, Mistral, Kimi	GPT-5, Claude, Gemini

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Closed vs Open Models

A lot of human and societal implications:

Privacy, ownership, safety, economics, control…

(we will return to those later)

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This Unit

Open and close models

What data do these models actually use?

How are the models evaluated?

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Model data and computation needs

What data do these models use?

Why is it challenging?

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What's Actually in the Training Data?

Pretraining data is a huge snapshot of the public web (and more)

Example: GPT-3 (OpenAI, 2020)

175B	~500B	~300B
parameters in the trained model	tokens in the filtered source pool	tokens used during training

Source: Brown et al., "Language Models are Few-Shot Learners" (2020).

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What's Actually in the Training Data?



GPT-3's training set: Roughly equivalent to three million books.

~300B

tokens used during training

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Where did that Data Come From?

Example: GPT-3 (2020) pretraining mix



- **Common Crawl (60%)** — Broad scrape of the public web, filtered for quality
- **WebText2 (22%)** — Curated web pages, expanded from GPT-2's Reddit-linked dataset
- **Books1 (8%)** — Internet-sourced book corpus
- **Books2 (8%)** — A second, larger internet-sourced book corpus
- **Wikipedia (3%)** — English-language Wikipedia articles

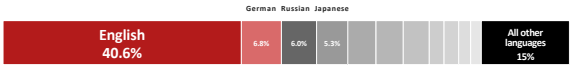
Source: Brown et al., "Language Models are Few-Shot Learners" (2020).

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Where did that Data REALLY Come From?

- **Common Crawl (2026 numbers)** — Broad scrape of the public web, filtered for quality



Source: Common Crawl

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Where did that Data REALLY Come From?

Common Crawl (2026 numbers)
Broad scrape of the public web, filtered for quality

English:	872,025,267 documents (40%)
Dzongkha (Bhutan):	1,996 documents (0.0001%)
Bengali (Bangladesh):	2,354,970 documents (0.11%)
Hindi:	4,766,382 (0.22%)
Spanish:	99,134,946 (4.6%)
Vietnamese:	22,749,990 (1.1%)
Tagalog (Philippines):	245,104 (0.01%)

Source: Common Crawl

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The other sources have even *more* English!

Example: GPT-3 (2020) pretraining mix



Common Crawl (60%)
Broad scrape of the public web, filtered for quality

WebText2 (22%)
Curated web pages, expanded from GPT-2's Reddit-linked dataset

Books1 (8%)
Internet-sourced book corpus

Books2 (8%)
A second, larger internet-sourced book corpus

Wikipedia (3%)
English-language Wikipedia articles

Source: Brown et al., "Language Models are Few-Shot Learners" (2020).

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English dominates all models

	English	Chinese
CommonCrawl	43.0	4.1
Llama 2	89.7	0.1
Llama 3.1	92.3	2.6
Qwen 3	73.4	22.8
Gemma 3	74.9	3.1

Estimates: Takase & Honda (2026)

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Training Data Size Over Time

Each new generation is trained on A LOT more data



GPT-2
~8 billion tokens

GPT-3
300 billion tokens

Llama 2
~2 trillion tokens

Llama 3
15 trillion tokens

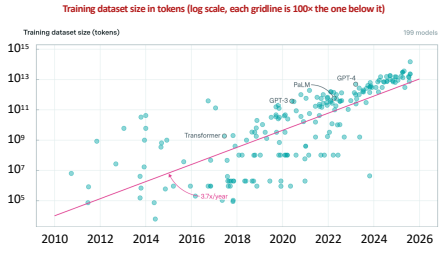
Sources: Brown et al. (2020); Touvron et al., LLaMA 2 (2023); Meta, "Introducing Meta Llama 3" (2024).

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Training Data Size Over Time

Training dataset size in tokens (log scale, each gridline is 100× the one below it)



Source: epoch.ai

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Data is King

More data → Can build larger models (more parameters) → Better performance

Companies no longer sharing what data they use:

"Claude Opus 5 was trained on a proprietary mix of publicly available information from the internet, public and private datasets, and synthetic data generated by other models."

— Claude Opus 5 System Card (Anthropic, 2026)

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Data is King… but Need Compute

More data → Can build larger models (more parameters) → Better performance

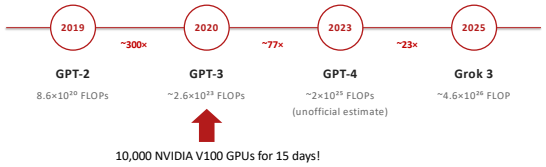
Larger models + more data → more computation required

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Compute Use Over Time

Training the largest models requires vastly more compute power



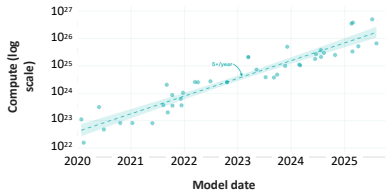
Sources: OpenAI compute analysis; SemiAnalysis (GPT-4 estimate, unconfirmed); Epoch AI, database of notable AI models (Grok 3, 2025).

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Compute Use Over Time

Training compute in FLOP — log scale, each gridline is 10× the one below



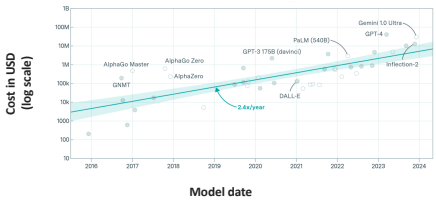
Source: epoch.ai; FLOP = floating-point operation, a single arithmetic step; total FLOP = how much computing went into training.

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Sidenote: huge costs

Model training compute cost (hardware and energy): log scale, each gridline is 10× the one below



Source: epoch.ai

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Problem: Getting More Data

To build larger models, companies need more and more data

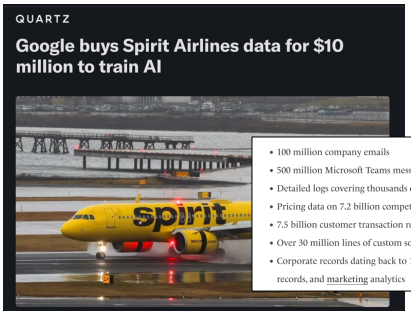
Already used pretty much all the Web (text, audio, video)

New Web content may not be human ("model collapse")

Raiding the archives: Books, movies, recordings…

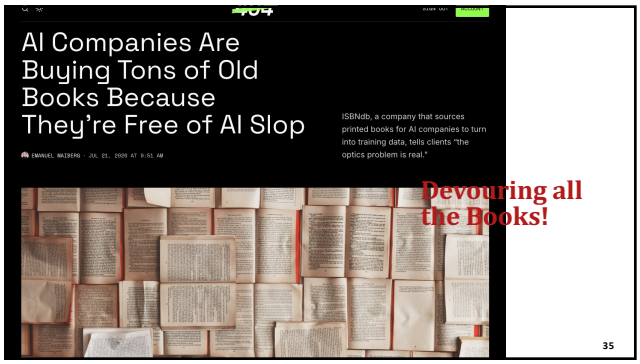
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Devouring all the Books!

- Providers like Anthropic, OpenAI buying books in bulk, scanning them to feed data for model training
- Pre-2022 best: human-written text
- Scanning usually destroys the book!
- **Raises a lot of questions:**
  - Do the authors need to be compensated?
    (more on that later)
  - Who decides if the physical books should be ruined?
  - What books are available? What languages?
  - Coming soon: bias in the selection of books and data sources

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Discussion Time!

- There is a library with 200,000 books in your language. It is the only copy of these books that is available.
- Would you give them to a company for AI/LLM training if the books will be destroyed during scanning?
- What if they pay you $10M? ($50 per book).
- Are there any reasons to give the books?

Take 5 minutes to discuss with the person next to you.

Submit your answer (reasons to give the books) here: pollev.com/profmor

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Summary

- More data, more compute → better models
- Data used for key models is mostly in English; few other language with good representation
- Companies devour all kinds of sources to get more data

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This Unit

Open and close models

What data do these models actually use?

How are the models evaluated?

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How Much Human Effort Goes In?

Making a model helpful takes real people, not just more data

Remember from earlier... help turn a LLM into a helpful chatbot:

Supervised Fine-Tuning
learn to follow instructions from examples

RLHF
learn what people actually prefer

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How Much Human Effort Goes In?

Making a model helpful takes real people, not just more data

Beyond those two, people are involved at every stage:

Data curation
choosing what goes into the training set, and writing the rules that filter it

Safety labeling
reading and tagging harmful content so a filter can learn to catch it

Red-teaming
people paid to break the model and find what it should not say

Evaluation
judging whether an answer is actually good, better, or simply wrong

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How Much Human Effort Goes In?

Making a model helpful takes real people, not just more data

Example: Claude Opus 5 (Anthropic, 2026)

> *"Anthropic partners with data work platforms to engage workers who help improve our models through **preference selection, safety evaluation, and adversarial testing**."*
>
> — Claude Opus 5 System Card (Anthropic, 2026)

preference selection → RLHF
safety evaluation → judging outputs
adversarial testing → red-teaming

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Human Effort: some problem

Who provides the human labor?

Much of it was done by low-paid raters in countries like Kenya, India, and Uganda

No safety, little support for work that is often problematic/traumatic.

preference selection → RLHF
safety evaluation → judging outputs
adversarial testing → red-teaming

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How Much Human Effort Goes In?

Example: InstructGPT (OpenAI, 2022) ← A first step between GPT 3.0 to ChatGPT!

~40 contractors hired through Upwork and ScaleAI, screened and trained on detailed guidelines

13K prompts with human-written demonstrations, used for supervised fine-tuning (SFT)

33K prompts with human-ranked comparisons of model outputs, used to train a reward model (RLHF)

Source: Ouyang et al., "Training language models to follow instructions with human feedback" (2022).

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How Much Human Effort Goes In?

Example: Llama 2 (Meta, 2023)

1.4M human preference comparisons collected by Meta with its own trained annotators

~40× more comparisons than InstructGPT

Source: Touvron et al., "Llama 2: Open Foundation and Fine-Tuned Chat Models" (2023).

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This Unit

Open and close models

What data do these models actually use?

How are the models evaluated?

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How do we measure model performance? How do we know how well they "work"?

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What we have seen before

Whether people like the answers

RLHF / RLVR

But what are the models eventually tested on?

How do we "know" that one model is better than others?

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Solution: benchmarks

Benchmarks are standardized exams for comparing models

Each has a fixed set of questions with known correct answers

Every model takes the same test, so scores are comparable

A single number (% correct) summarizes performance on that skill

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Testing Without Peeking

Benchmark answers are kept out of training data so scores are meaningful



Held-Out Answers

The correct answers used for scoring aren't published in a form crawlers can easily find.

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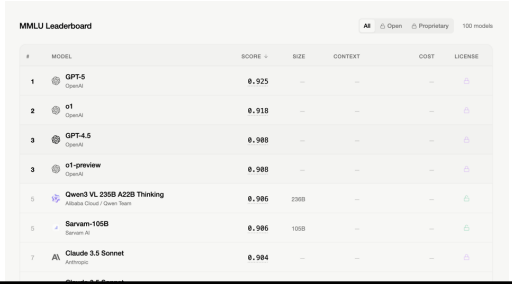
Example benchmarks

MMLU	knowledge exam with 57 subjects (details next)
HumanEval	checks whether generated code actually runs correctly
GSM8K	grade-school math word problems that test reasoning
HellaSwag	picks the most sensible ending to an everyday scenario
TruthfulQA	checks model resistance to common misconceptions

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Benchmarks leaderboard



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Example: MMLU (2021)

Massive Multitask Language Understanding

A general knowledge-and-reasoning leaderboard for models, Covers everything from elementary school to professional exams

57	~16,000	4
subjects	*questions*	*answers (multiple choice)*

Source: Hendrycks et al., "Measuring Massive Multitask Language Understanding" (2021).

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MMLU in Practice

An example, in MMLU's style

Subject: High School Chemistry

"A water solution has a pH of 3. Which best describes it?"

A) Strongly basic
B) Weakly basic
C) Neutral
D) Acidic

Correct answer: **D**

See https://huggingface.co/datasets/cais/mmlu/viewer/ for real questions

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MMLU in Practice

An example, in MMLU's style

Subject: World History

"The 1884–85 Berlin Conference was convened primarily to regulate the partition of which continent among European powers?"

A) South America
B) Africa
C) Southeast Asia
D) The Middle East

Correct answer: **B**

See https://huggingface.co/datasets/cais/mmlu/viewer/ for real questions

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Benchmarks: not perfect

Models get optimized to specific tasks (who gets to pick the tasks?)

Biased coverage

Saturation: top models already near 100%, little room to improve!

Gaming the test: training on similar data can inflate scores

Contamination and leakage

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Benchmarks: not perfect

Not real: users may use different framing, wording that will effect results

No context --- models may act differently depending on user history, chat history, etc.

"Real" evaluation with users and their prompts is hard!

Kapoor, S., et al. (2026). Open-World Evaluations for Measuring Frontier AI Capabilities.

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Summary: benchmark evaluations

- Benchmarks are standardized exams, so every model can take the same test
- The measure (% correct answers) makes models directly comparable
- Many Challenges!
- "Real" evaluation with actual users, actual tasks is hard

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Bangla Benchmarks?

There are several Bangla benchmarks, for example:

"We present BnMMLU, a comprehensive benchmark for measuring massive multitask language understanding in Bengali. BnMMLU spans 41 domains across STEM, humanities, social sciences, and general knowledge, and contains 134,375 multiple-choice question–option pairs-the most extensive Bengali evaluation suite to date."

Source Joy and Shatabda (2026) :BnMMLU: Measuring Massive Multitask Language Understanding in Bengali

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Exercise: Create a Benchmark

To understand the challenges and issues that are involved in creating a benchmark, we will create a Bangladesh knowledge benchmark and test it in English/Bangla.

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Exercise: Create a Benchmark, Step 1

In groups of 3-4 people, create a Bangladesh Benchmark of at least 10 questions about Bangladesh.

The questions should be:

1) With multiple choice answers
2) Not easily answered on the Internet
3) Hard, but answerable by a human expert or country native

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Exercise: Create a Benchmark, Step 2

With your questions, test two models of your choice (for example, ChatGPT and DeepSeek) to see how many questions they get right.

Ask the two models your 10 questions, one after the other, using the format below (replace "Question" and "Answer" with your question and answers). Then do the same in Bangla.

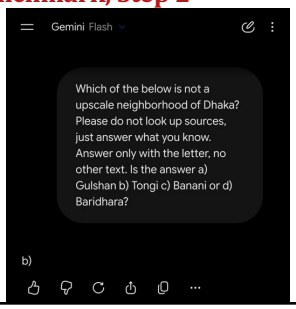
<QUESTION>. Please do not look up sources, just answer what you know. Answer only with the letter, no other text. Is the answer a) <ANSWER A> b) <ANSWER B> c) <ANSWER C> or d) <ANSWER D>?

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Exercise: Create a Benchmark, Step 2

Example:



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Exercise: Create a Benchmark, Step 2

With your questions, test two models of your choice (for example, ChatGPT and DeepSeek) to see how many questions they get right in English and Bangla. Start a new chat for every question. Make sure the chat does not look up the answer (sometimes they do even when requested not to). Count the correct answers for each model/language.

Submit the results using "Benchmark Results" link at ai-course.mornaaman.com

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Exercise: Create a Benchmark, step 3

Upload 2-3 of your best questions here (one question at a time)

Best to choose questions that at least one of the models answered incorrectly!

pollev.com/profmor

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Exercise: Create a Benchmark, step 4

Vote on the best questions in your opinion!

pollev.com/profmor

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Summary: this Unit

Open and close models

- Closed-weight (GPT-5, Claude, Gemini) = API-only, weights private
- Open-weight (Llama, DeepSeek, Qwen…) = downloadable, so anyone can run or fine-tune it

What data do these models actually use?

- Massive web scrapes plus books and more, English dominates

How are the models evaluated? (benchmarks)

- Benchmarks (MMLU, HumanEval, GSM8K…) for models, but suffer contamination, gaming, and saturation

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